Whitehead Field Dynamics

Computational Notes

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Abstract

This companion note turns the executable notebook of the same name into a short transcript. It records the local pair-averaging calculation, complete graph-of-graphs models through $N = 8$, two rules for choosing the next graph type, an exact 71-state diffusion calculation, one seeded 48-step path, and finite consensus products for K_4 and Q_3. Exact rational arithmetic is used until a decimal display is requested. The proofs are given in the full manuscript.

1 What is being computed

A *cubic multigraph* is a graph in which every vertex has degree three; a loop contributes two to the degree, and parallel edges are allowed. A *graph type* is an isomorphism class obtained after vertex labels are forgotten. The *graph of graphs* (GoG) has one vertex for each connected graph type and joins two types when one Whitehead move relates them.

A Whitehead move selects a *carrier edge*, meaning a specific edge in a chosen representative, collapses it, and chooses one of the two other pairings of the four exposed half-edges. A scalar field $f \in \mathbb{R}^N$ travels with the graph. If the carrier has endpoints i, j, the field update is

$$(f_i, f_j) \longmapsto \left(\frac{f_i + f_j}{2}, \frac{f_i + f_j}{2} \right).$$

Three computational objects must be kept separate.

1. The carrier field uses the concrete selected edge. After an unlabelled move, a *settlement permutation* is the relabelling that identifies the result with the stored representative of its graph type; the same permutation acts on the field coordinates.
2. Rule A keeps every local move. A cubic graph has $3N/2$ edges and two resolutions per edge, hence $3N$ moves. Different moves leading to the same graph type retain their multiplicity. A move returning to the current type is a *holding event*.
3. Rule B assigns probability $1/(3N)$ to each distinct adjacent graph type and the remaining probability to staying put. Its *jump chain*, denoted `"BJump"`, is the conditional chain given that the graph type changes. Removing holds changes path probabilities and elapsed time.

The notebook bundles every graph type, every one-move connection, and the exact transition counts for $N = 2, 4, 6, 8$. The $N = 8$ model has 71 states; larger data sets can be generated separately.

2 Notebook map

The computational sections of the executable companion are:

Notebook section	Computational role
Graph of graphs	Shows the five $N = 4$ types, then provides a live inspector for the complete $N = 2, 4, 6, 8$ models, including all 71 vertices at $N = 8$.
Local averaging	Turns one pair average into the exact loss $(f_i - f_j)^2/2$.
Bundled models	Checks class counts, the GoG edge list, transition-row sums, Rule-B symmetry, and preservation of the uniform distribution.
Five-state example	Displays the complete Rule-A count matrix and Rule-B transition matrix at $N = 4$ in exact arithmetic.
Diffusion	Places the exact time-12 distribution and a separate 48-step sample on the 71-state $N = 8$ GoG.
Finite-time consensus	Verifies the averaging products for K_4 and Q_3.

The notebook keeps explanatory prose short and places executable cells beside the relevant picture. In this note, every input/output pair has one shared caption of the form "Mathematica cells: *topic*." Input cells retain the pale-blue background used in the notebook, while exact outputs retain the warm cream background. This makes the direction of a calculation visible without repeating labels such as "input" and "output" above every box.

3 The local averaging calculation

For a non-loop pair $\{i, j\}$, the averaging matrix is

$$A_{ij} = I - \frac{1}{2}(e_i - e_j)(e_i - e_j)^T.$$

It is a symmetric idempotent matrix, fixes $\mathbf{1}$, and hence is the orthogonal projection onto $\{f : f_i = f_j\}$. The two-coordinate energy calculation can be checked without constructing the whole matrix.

Mathematica cells: one-pair loss

```
Clear[x, y];
FullSimplify[
  x^2 + y^2 - 2 ((x + y)/2)^2,
  Assumptions -> Element[{x, y}, Reals]
]
```

```
(x - y)^2/2
```

Since the sum of the two coordinates is unchanged, the mean is unchanged as well. Applying the identity to the centered field gives the exact variance identity

$$\mathcal{V}(A_{ij}f) = \mathcal{V}(f) - \frac{1}{2}(f_i - f_j)^2, \qquad \mathcal{V}(f) = \|f - \bar{f}\mathbf{1}\|_2^2.$$

The following definition is used again for the consensus calculation.

Mathematica cells: averaging matrix

```
pairAverage[n_, {i_, j_}] := ReplacePart[
  IdentityMatrix[n],
  {{i, i} -> 1/2, {i, j} -> 1/2,
   {j, i} -> 1/2, {j, j} -> 1/2}
];

a = pairAverage[4, {1, 3}];
{
  SymmetricMatrixQ[a],
  a . a === a,
  a . ConstantArray[1, 4] === ConstantArray[1, 4]
}
```

```
{True, True, True}
```

A settlement after labels are forgotten is a permutation matrix P_σ. Because P_σ preserves sums and Euclidean norms, replacing A_{ij} by $P_\sigma A_{ij}$ does not change this one-step variance identity. It does, however, affect which persistent pair of field coordinates is averaged later. A list of unlabelled graph types therefore does not contain enough information to reconstruct the later field values.

4 Loading and validating the notebook models

From a notebook opened at the repository root, the local companion package can be loaded as follows.

Mathematica cells: local package

```
repo = NotebookDirectory[];
PacletDirectoryLoad[
  FileNameJoin[{repo, "Wolfram", "GogFieldDynamics"}]
];
Needs["GogFieldDynamics`"];

GogAvailableSizes[]
```

```
{2, 4, 6, 8}
```

For each bundled size, the next query records

$$\{N,\ \text{number of classes},\ \text{GoG edges},\ \text{total Rule-A moves}\}.$$

Mathematica cells: resource summary

```
Table[
  With[{m = GogSmallModel[n]},
    {n, m["NodeCount"], m["EdgeCount"],
     m["RuleASlotCount"]}
  ],
  {n, GogAvailableSizes[]}
]
```

```
{{2, 2, 1, 12}, {4, 5, 4, 60}, {6, 17, 25, 306},
 {8, 71, 179, 1704}}
```

The class counts $2, 5, 17, 71$ agree with the corresponding terms of OEIS A005967. The last column satisfies the independent count

$$\#\text{classes} \times 3N: \qquad 2 \cdot 6 = 12, \quad 5 \cdot 12 = 60, \quad 17 \cdot 18 = 306, \quad 71 \cdot 24 = 1704.$$

The next cell checks that every bundled table is complete, that the node and outcome counts agree, and that every transition row sums to one. It also checks Rule-B symmetry and verifies that a uniform starting distribution remains uniform after one step.

Mathematica cells: compact validation report

```
r = GogValidateSmallModels[];
{
  r["Valid"],
  AssociationMap[
    r["Models", #]["Valid"] &,
    GogAvailableSizes[]
  ]
}
```

```
{True, <|2 -> True, 4 -> True, 6 -> True, 8 -> True|>}
```

These checks validate the bundled finite data and the implemented matrix conventions. They are not an asymptotic theorem.

5 An exact five-state example

At $N = 4$ there are five unlabelled graph types. Rule A counts every local move outcome, including repeated outcomes that lead to the same graph type. Its count matrix is

Mathematica cells: Rule-A counts

```
countsA = GogTransitionCounts[4, "A"];
{countsA, Total /@ countsA}
```

```
{{{4, 4, 4, 0, 0},
  {6, 6, 0, 0, 0},
  {2, 0, 8, 0, 2},
  {0, 0, 0, 6, 6},
  {0, 0, 4, 2, 6}},
 {12, 12, 12, 12, 12}}
```

Dividing by $3N = 12$ gives the exact transition matrix

$$P_A = \begin{pmatrix} 1/3 & 1/3 & 1/3 & 0 & 0 \\ 1/2 & 1/2 & 0 & 0 & 0 \\ 1/6 & 0 & 2/3 & 0 & 1/6 \\ 0 & 0 & 0 & 1/2 & 1/2 \\ 0 & 0 & 1/3 & 1/6 & 1/2 \end{pmatrix}.$$

This matrix is not symmetric. A *stationary distribution* is a probability row vector π satisfying $\pi P = \pi$. Solving that equation for Rule A with exact arithmetic gives

Mathematica cells: Rule-A stationary law

```
pA = GogTransitionMatrix[4, "A"];
piA = First[
  NullSpace[Transpose[pA] - IdentityMatrix[Length[pA]]]
];
piA/Total[piA]
```

```
{1/5, 2/15, 2/5, 1/15, 1/5}
```

The vector is nonuniform because graph types with different numbers of labelled realizations receive different weights. Treating Rule A as uniform on graph types would therefore change the process.

Under Rule B, every different graph type reachable in one move has probability $1/12$. Whatever probability remains is assigned to staying at the current graph type:

Mathematica cells: Rule B probabilities at $N = 4$

```
pB = GogTransitionMatrix[4, "B"];
MatrixForm[pB]
```

```
{{5/6, 1/12, 1/12, 0, 0},
 {1/12, 11/12, 0, 0, 0},
 {1/12, 0, 5/6, 0, 1/12},
 {0, 0, 0, 11/12, 1/12},
 {0, 0, 1/12, 1/12, 5/6}}
```

The defining structural checks are visible directly in the output:

Mathematica cells: symmetry and stationarity

```
u = ConstantArray[1/5, 5];
{SymmetricMatrixQ[pB], u . pB}
```

```
{True, {1/5, 1/5, 1/5, 1/5, 1/5}}
```

For a symmetric transition matrix, the *spectral gap* is one minus its second-largest eigenvalue. Here it is exact:

Mathematica cells: Rule-B spectrum and gap

```
eigenvaluesB = Sort[Eigenvalues[pB], Greater];
{eigenvaluesB,
 FullSimplify[1 - eigenvaluesB[[2]]]}
```

```
{{1, (21 + Sqrt[5])/24, (19 + Sqrt[5])/24,
  (21 - Sqrt[5])/24, (19 - Sqrt[5])/24},
 (3 - Sqrt[5])/24}
```

Thus Rule B with its stay-put steps retained is symmetric, and the uniform class law is stationary. The *support graph* keeps one edge whenever a nonself transition is possible and discards transition multiplicities. If L_{supp} is its graph Laplacian, then

$$I - P_B = \frac{1}{3N} L_{\mathrm{supp}}.$$

Staying in place is part of the random process and cannot simply be discarded. Already at $N = 2$, Rule B and the chain conditioned on moving are

Mathematica cells: Rule B versus its jump chain

```
{
  GogTransitionMatrix[2, "B"],
  GogTransitionMatrix[2, "BJump"]
}
```

```
{{{5/6, 1/6}, {1/6, 5/6}},
 {{0, 1}, {1, 0}}}
```

The two matrices allow the same changes to other graph types, but have different spectra and path probabilities. A holding step consumes one Rule-B step; the jump chain conditions on such a hold not occurring.

6 Exact diffusion and a sampled path

Here *diffusion* means the evolution of a probability distribution over graph types. From a point mass at $G1$,

$$p_t = e_{G1}^T P^t, \qquad p_{t+1} = p_t P, \qquad \sum_g p_t(g) = 1.$$

The following five-state calculation keeps every rational entry visible.

Mathematica cells: default exact snapshot

```
p4A = GogDiffusionState[4, "A", 1, 4];
{p4A, N[p4A, 6]}
```

```
{{185/648, 49/216, 59/162, 1/54, 17/162},
 {0.285494, 0.226852, 0.364198, 0.0185185, 0.104938}}
```

All five classes are reachable by time four, but the mass is far from the stationary vector. From the same starting point, Rule B retains much more probability at the first class because it more often stays in place:

Mathematica cells: Rule-B comparison

```
p4B = GogDiffusionState[4, "B", 1, 4];
{p4B, N[p4B, 6]}
```

```
{{5623/10368, 1181/5184, 4121/20736,
  41/20736, 151/5184},
 {0.542342, 0.227816, 0.198736, 0.00197724, 0.0291281}}
```

The extra stay-put probability in Rule B is plainly visible: after four steps, more than half of the probability remains at the starting graph type. These calculations compare how Rule A and Rule B move probability among graph types. They do not yet describe how a scalar field moves under Rule B: when several edge moves lead to the same next graph type, one must still say which actual edge is averaged.

The notebook then repeats the calculation on the complete $N = 8$ GoG. Blue encodes the exact time-12 distribution on all 71 graph types; the orange trail is a separate 48-step sample and is not used to estimate the blue values.

Mathematica cells: 71-state diffusion and path summary

```
p8A = GogDiffusionState[8, "A", 1, 12];
path8 = GogRandomTrajectory[8, "A", 1, 48, 273];

{
  {Length[p8A], Count[p8A, _?Positive], Total[p8A],
   N[Max[p8A], 7]},
  {Length[path8] - 1, Length[DeleteDuplicates[path8]],
   Count[Partition[path8, 2, 1], {x_, x_}]}
}
```

```
{{71, 71, 1, 0.06504625}, {48, 27, 13}}
```

Thus every $N = 8$ graph type has positive mass by time 12. The sample visits 27 distinct types; its 13 repeated consecutive labels are genuine holding events. Rule-A multiplicity is retained because sampling uses the exact integer count matrix.

7 Finite-time consensus

Consensus means that every field coordinate equals the arithmetic mean of the initial field: $f \mapsto \bar{f}\mathbf{1}$. *Finite-time consensus* means that a finite product of pair-averaging matrices equals

$$J_N = \frac{1}{N}\mathbf{1}\mathbf{1}^T, \qquad J_N f = \bar{f}\mathbf{1}.$$

A *perfect matching* is a set of disjoint pairs covering all N coordinates. Pair averages within one perfect matching commute, so the matching can be treated as one block. The following calculation checks two blocks on K_4 and three coordinate matchings on the cube Q_3.

Mathematica cells: matching products

```
matching[n_, edges_] :=
  Dot @@ (pairAverage[n, #] & /@ edges);

k4Product =
  matching[4, {{1, 3}, {2, 4}}] .
  matching[4, {{1, 2}, {3, 4}}];

q3Product =
  matching[8, {{1, 5}, {2, 6}, {3, 7}, {4, 8}}] .
  matching[8, {{1, 3}, {2, 4}, {5, 7}, {6, 8}}] .
  matching[8, {{1, 2}, {3, 4}, {5, 6}, {7, 8}}];

{
  k4Product === ConstantArray[1/4, {4, 4}],
  q3Product === ConstantArray[1/8, {8, 8}]
}
```

```
{True, True}
```

For example, the first product is returned exactly as

Mathematica cells: the K_4 product

```
k4Product // MatrixForm
```

```
{{1/4, 1/4, 1/4, 1/4},
 {1/4, 1/4, 1/4, 1/4},
 {1/4, 1/4, 1/4, 1/4},
 {1/4, 1/4, 1/4, 1/4}}
```

Consequently every initial field reaches its exact mean after the displayed schedule. This is stronger than exponential convergence, and it explains the exceptional finite-time consensus at $N = 4$ and $N = 8$. It should not be extrapolated to arbitrary N: every entry of a finite product of pair averages has denominator a power of two, whereas the mean of $(0, 0, 1)$ requires the factor $1/3$.

References

[1] Laura Grave de Peralta and Alexander Kolpakov, *Expansion properties of Whitehead moves on cubic graphs*, Comptes Rendus Mathématique **362** (2024), 1825–1836. https://doi.org/10.5802/crmath.691

[2] Michael Li, *Increasingly Many Bounded Eigenvalues of the Graph of Whitehead Moves*, arXiv:2405.14592. https://arxiv.org/abs/2405.14592